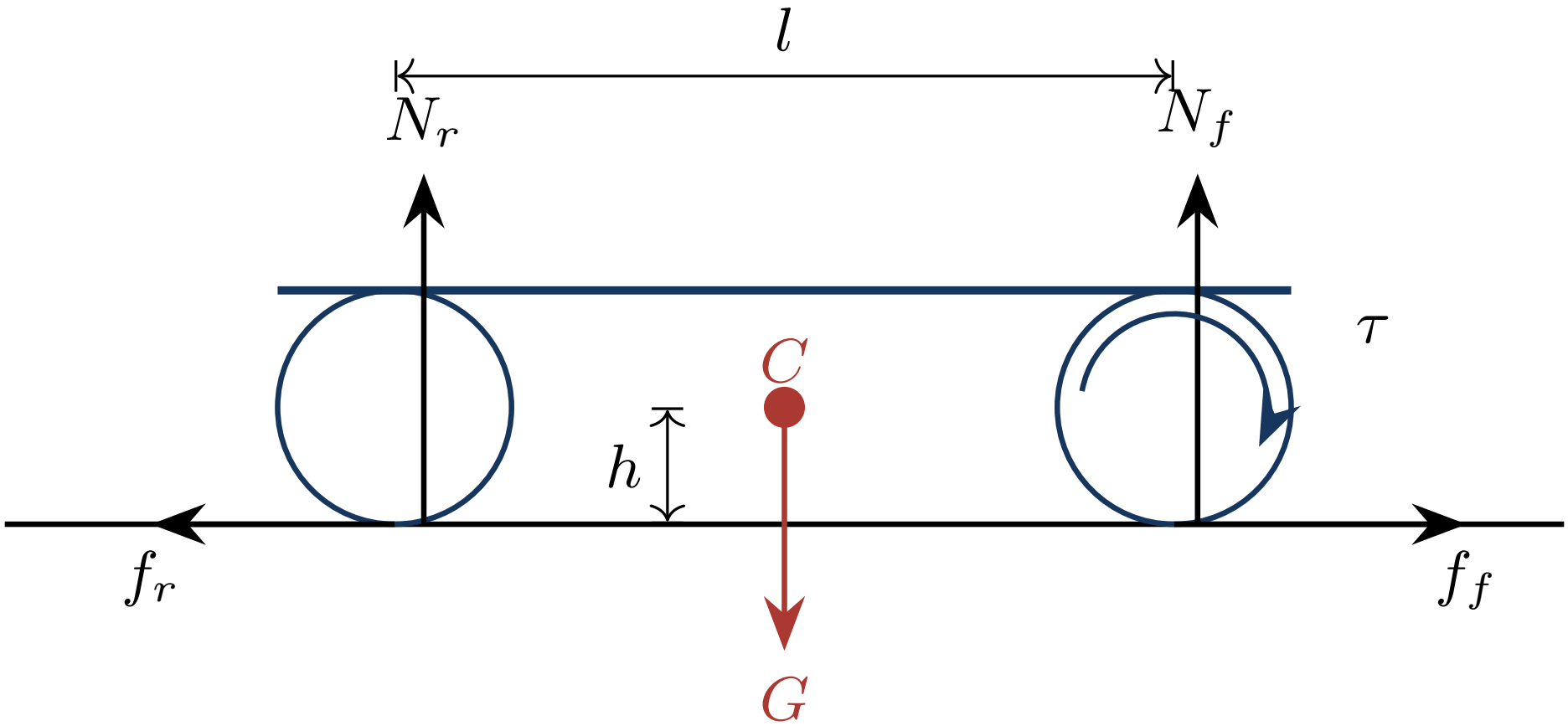
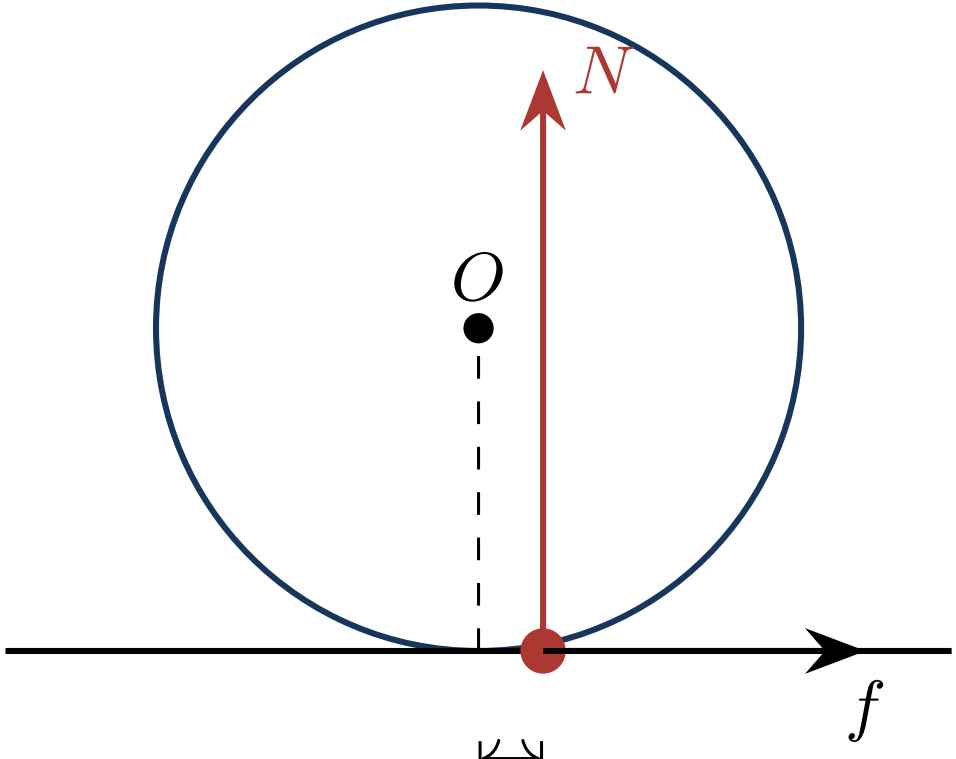
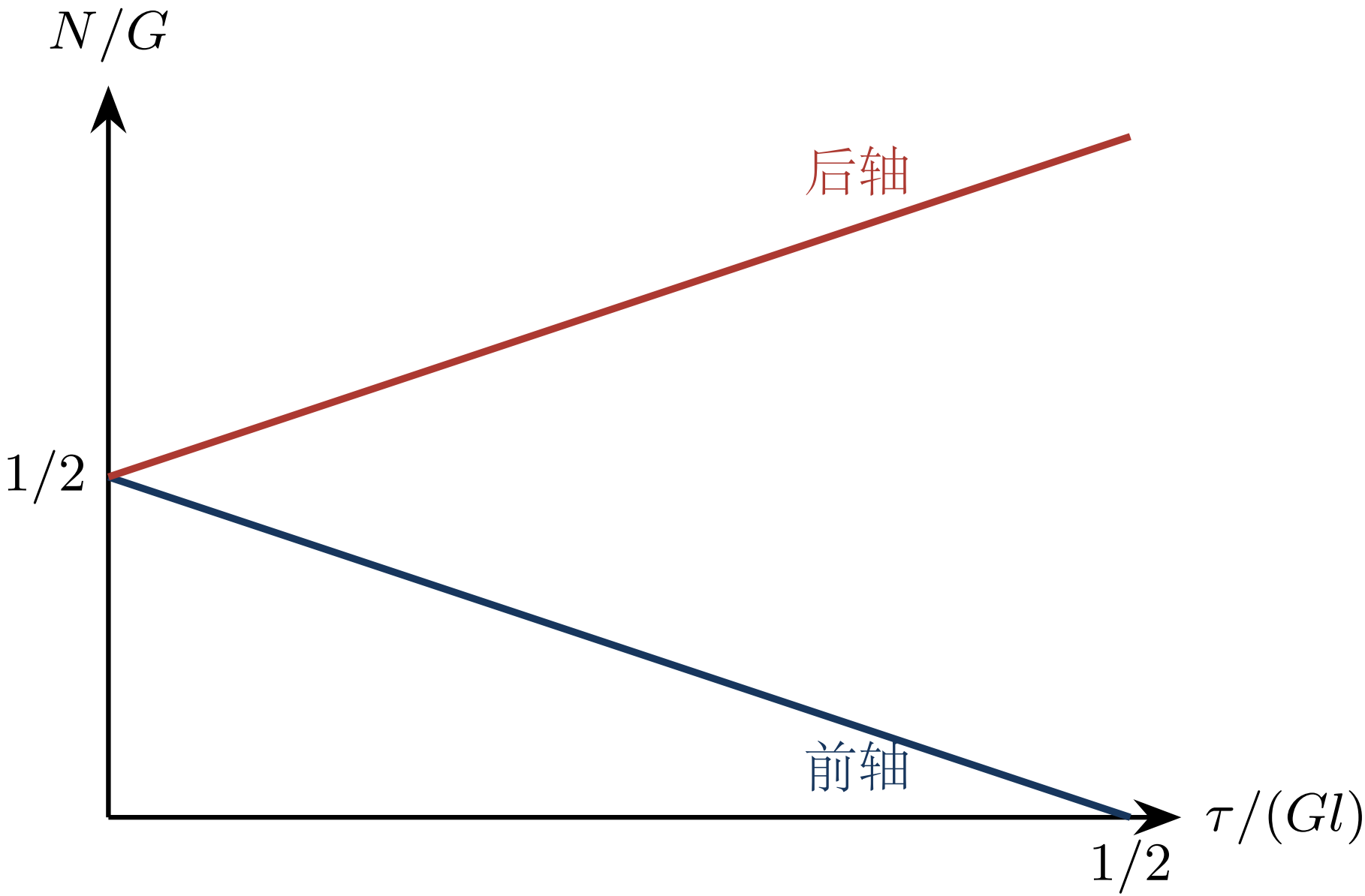

l
N_r
N_f
τ
C
h
f_r
f_f
G
后轴
前轴（驱动）



前轮：$hf = \tau - Nd$

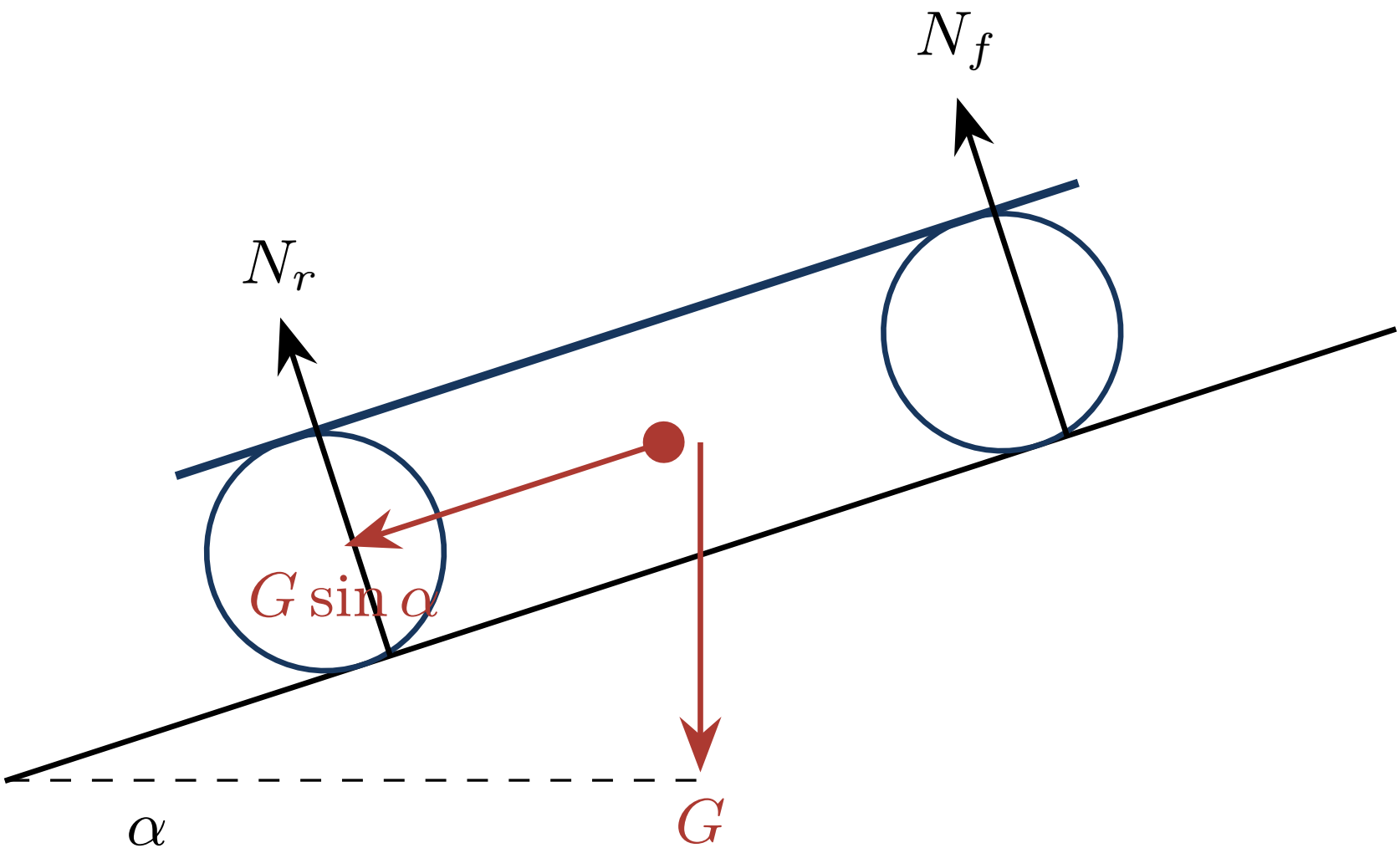
后轮：$hf = -Nd$

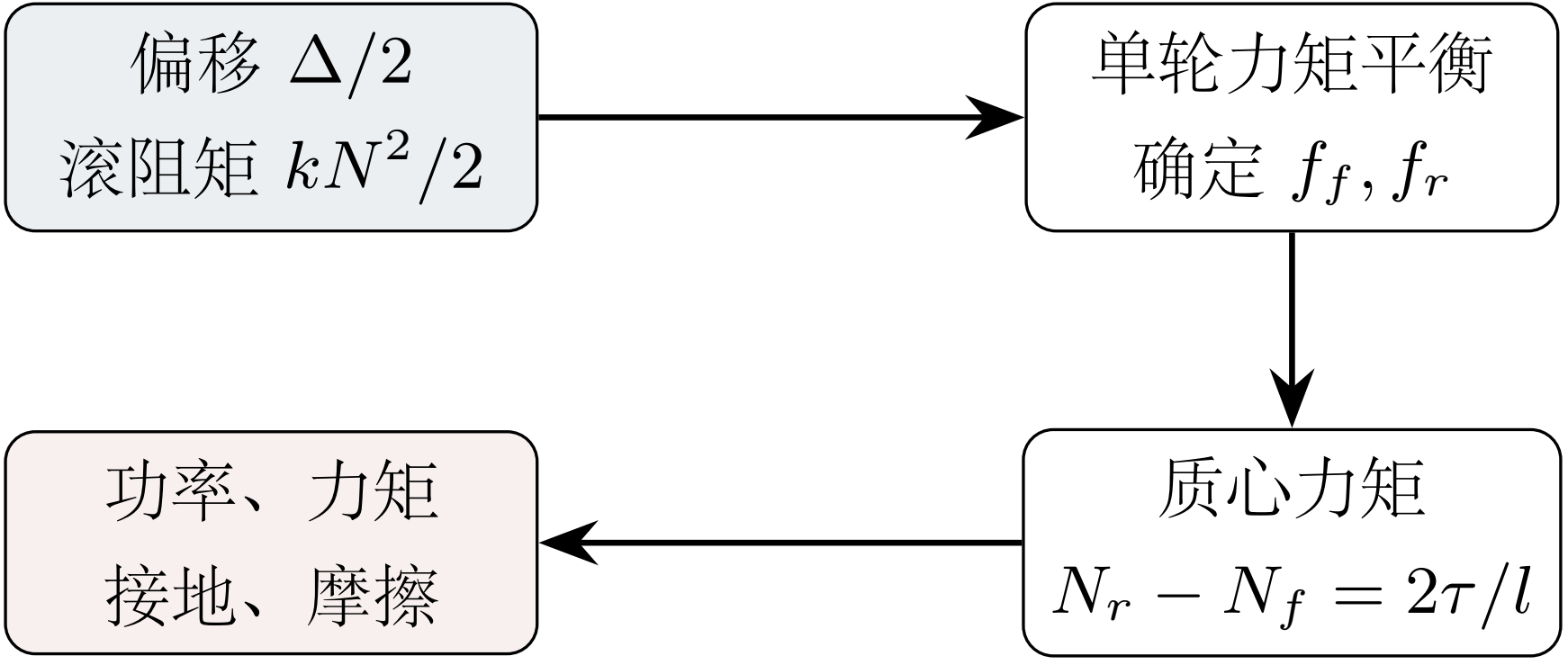
$d = kN/2$



$N_f > 0$ 限定双轴接地范围

$$N_f + N_r = G\cos\alpha$$




偏移 $\Delta/2$
滚阻矩 $kN^2/2$
单轮力矩平衡
确定 f_f, f_r
质心力矩
$N_r - N_f = 2\tau/l$
功率、力矩
接地、摩擦